Differentiation

ISC Class 12 Previous Years Board questions

with Answers

2000 to Sem1-2021

1. $If\ \ y = \log\sqrt{\frac{1-\cos x}{1+\cos x}},\ find\ \ \frac{dy}{dx}$ $(isc\ \ 2000)$

2. $If\ \ y = (\cos x)^{\cos x}, find\ \ \frac{dy}{dx}$ $(isc\ \ 2001)$

3. $If\ \ y = e^x \log(tan\ 2x),\ \ find\ \frac{dy}{dx}$ $(isc\ \ 2002)$

4. $If\ \ y = \tan^{-1}\left(\frac{2x}{1-x^2}\right),\ prove\ that\ \ \frac{dy}{dx} = \frac{2}{1+x^2}$ $(isc\ \ 2003)$

5. $If\ \ y = e^{m\cos^{-1}x},\ prove\ that\ (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2y$ $(isc\ \ 2003)$

6. $If\ \ x^y y^x = 5,\ show\ that\ \ \frac{dy}{dx} = -\left(\frac{\log y + \frac{y}{x}}{\log x + \frac{x}{y}}\right)$ $(isc\ \ 2004)$

7. $If\ \ x = a\ sin^3 t\ \ and\ \ y = a\ cos^3 t,\ \ find\ \ \frac{dy}{dx}$ $(isc\ \ 2004)$

8. $If\ \ \sin(xy) + \cos(xy) = 1\ \ and\ \ \tan(xy) \neq 1, then\ show\ that\ \ \frac{dy}{dx} = -\frac{y}{x}$ $(isc\ \ 2005)$

9. $If\ \ x^p y^q = (x+y)^{p+q}, prove\ that\ \ \frac{dy}{dx} = \frac{y}{x}$ $(isc\ \ 2005)$

10. $If\ \ y = e^{\sin x^2},\ \ find\ \ \frac{dy}{dx}$ $(isc\ \ 2006)$

11. $If\ \ y = \frac{\sin^{-1}x}{\sqrt{1-x^2}}, prove\ that\ (1-x^2)\frac{dy}{dx} - xy = 1$ $(isc\ \ 2006)$

12. $If\ e^{x+y} = xy\ , show\ that\ \frac{dy}{dx} = \frac{y(1-x)}{x(y-1)}$ $(isc\ 2007)$

13. $If\ \sin y = x\ \sin(a+y)\ , show\ that\ \frac{dy}{dx} = \frac{sin^2(a+y)}{\sin a}$ $(isc\ 2007)$

14. $Find\ \frac{dy}{dx}\ if\ y = \tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ $(isc\ 2008)$

15. $If\ y = \sqrt{\frac{1-\cos x}{1+\cos x}}\ , find\ \frac{dy}{dx}$ $(isc\ 2008)$

16. $Using\ a\ suitable\ substitution\ , find\ the\ derivative\ of\ \tan^{-1}\frac{4\sqrt{x}}{1-4x}\ with\ respect\ to\ x$ $(isc\ 2009)$

17. $Find\ the\ derivative\ of\ \sin x^2\ with\ respect\ to\ x^3$ $(isc\ 2009)$

18. $Using\ a\ suitable\ substitution\ , find\ the\ derivative\ of\ \tan^{-1}\sqrt{\frac{a-x}{a+x}}\ with\ respect\ to\ x$

$(isc\ 2010)$

19. $If\ y = x^x, prove\ that\ \frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$ $(isc\ 2011)$

20. $If\ e^y(x+1) = 1\ , then\ show\ that\ \frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$ $(isc\ 2012)$

21. $If\ y = (\cot^{-1}x)^2\ , show\ that\ (1+x^2)^2\frac{d^2y}{dx^2} + 2x(1+x^2)\frac{dy}{dx} = 2$ $(isc\ 2013)$

22. $If\ y = \frac{x\ \sin^{-1}x}{\sqrt{1-x^2}}\ , prove\ that\ (1-x^2)\frac{dy}{dx} = x + \frac{y}{x}$ $(isc\ 2014)$

23. $If\ y = e^{m\cos^{-1}x}, prove\ that\ (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2y$ $(isc\ 2015)$

24. $If\ \log y = \tan^{-1}x\ , prove\ that\ (1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$ $(isc\ 2016)$

25. $If\ y = \cos(\sin x)\,, show\ that\ \frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\ cos^2 x = 0$ $(isc\ 2017)$

26. $If x = \tan\left(\frac{1}{a}\log y\right)\,, prove\ that\ (1+x^2)\frac{d^2y}{dx^2} + (2x-a)\frac{dy}{dx} = 0$ $(isc\ 2018)$

27. $If\ y = e^{\sin^{-1}x}\ and\ z = e^{-\cos^{-1}x}\,, prove\ that\ \frac{dy}{dz} = e^{\pi/2}$ $(isc\ 2019)$

28. $Find\ \frac{dy}{dx}\ if\ x^3 + y^3 = 3axy$ $(isc\ 2020)$

29. $If\ y = e^{m\sin^{-1}x},\ prove\ that\ (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2\,y$ $(isc\ 2020)$

30. Select and write the correct option: $(isc\ Sem\ 1-2021)$

Differentiation of $\log(1+x^2)$ with respect to $\tan^{-1}x$ is:

(a) $\frac{1}{1+x^2}$
(b) $2x$
(c) $\frac{-1}{1+x^2}$
(d) $-2x$

Answer: _____

31. Select and write the correct option: $(isc\ Sem\ 1-2021)$

If $x^2 + y^3 = 42$, then $\frac{dy}{dx}$ is:

(a) $\frac{dy}{dx} = \frac{-3\,y^2}{2x}$
(b) $\frac{dy}{dx} = \frac{3\,y^2}{2\,x}$
(c) $\frac{dy}{dx} = \frac{2\,x}{3\,y^2}$
(d) $\frac{dy}{dx} = \frac{-2\,x}{3\,y^2}$

Answer: ----------

32. Select and write the correct option: $(isc\ Sem\ 1 - 2021)$

If $x = a\ \cos\theta$, $y = a\ \sin\theta$, then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$ will be:

(a) $\frac{dy}{dx} = -1$

(b) $\frac{dy}{dx} = 1$

(c) $\frac{dy}{dx} = 0$

(d) $\frac{dy}{dx} = 2$

Answer: _____

33. Select and write the correct option: $(isc\ Sem\ 1 - 2021)$

If $y = \tan^{-1} x$, then

(a) $(1 + x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(b) $\sqrt{1 - x^2}\ \frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(c) $(1 - x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(d) $\sqrt{1 + x^2}\ \frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

Answer: _____

Answer

(1)	$cosec\ x$	(2)	$-(\cos x)^{\cos x}\ \sin x\ \ \log(e\cos x)$
(3)	$e^x\{2\ \cot 2x\ \ sec^2 2x + \log(\tan 2x)\}$	(7)	$-\ \cot t$
(10)	$2x\ \cos(x^2)\ e^{\sin(x^2)}$	(14)	$\frac{1}{2}\left(\frac{1}{1+x^2}\right)$
(15)	$\frac{1}{2}sec^2\left(\frac{x}{2}\right)$	(16)	$\frac{2}{(1+4x)\sqrt{x}}$
(17)	$\frac{2\ \cos(x^2)}{3x}$	(18)	$\frac{-1}{2\sqrt{a^2 - x^2}}$
(28)	$\frac{ay - x^2}{y^2 - ax}$	(30)	(b)
(31)	(d)	(32)	(c)
(33)	(a)		

Differentiation

Solution of

Previous Years ISC Class 12 Board questions

2000 to Sem1-2021

1. $If\ y = \log\sqrt{\frac{1-\cos x}{1+\cos x}}, find\ \frac{dy}{dx}.$ $(isc\ 2000)$

Solution: Given $y = \log\sqrt{\frac{1-\cos x}{1+\cos x}}$

$$= \log\sqrt{tan^2\left(\frac{x}{2}\right)}$$

$$= \log\left(\tan\frac{x}{2}\right)$$

$$\therefore \frac{dy}{dx} = \frac{1}{\tan\left(\frac{x}{2}\right)}.\frac{1}{2}.sec^2\left(\frac{x}{2}\right)$$

$$= \frac{1}{2}.\frac{\cos\left(\frac{x}{2}\right)}{\sin\left(\frac{x}{2}\right)}.\frac{1}{cos^2\left(\frac{x}{2}\right)}$$

$$= \frac{1}{2\ \sin\left(\frac{x}{2}\right)\ \cos\left(\frac{x}{2}\right)}$$

$$= \frac{1}{\sin x}$$

$$= \operatorname{cosec} x$$ Answer.

2. $If\ y = (\cos x)^{\cos x}, find\ \frac{dy}{dx}.$ $(isc\ 2001)$

Solution: Given $y = (\cos x)^{\cos x}$

Taking log both sides

$$\log y = \cos x\ \log(\cos x)$$

Differentiating each term $w.r.t.x$

$$\frac{1}{y}\frac{dy}{dx} = \cos x\ \frac{1}{\cos x}\ (-\sin x) + (-\sin x)\log(\cos x)$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = -\sin x - \sin x\ \log(\cos x)$$

$$\Rightarrow \frac{dy}{dx} = -y\ \sin x\ \{1 + \log(\cos x)\}$$

$$\Rightarrow \frac{dy}{dx} = -(\cos x)^x\ \sin x\ \{\log e + \log(\cos x)\}$$

$$\Rightarrow \frac{dy}{dx} = -(\cos x)^x\ \sin x\ \{\log(e\ \cos x)\}$$ Answer.

3. $If\ \ y = e^x\ \log(\tan 2x), find\ \frac{dy}{dx}.$ $(isc\ \ 2002)$

Solution: Given $y = e^x\ \log(\tan 2x)$

$$\Rightarrow \frac{dy}{dx} = e^x\ \frac{1}{\tan 2x}.2\ sec^2(2x) + e^x\ \log\tan 2x$$

$$\Rightarrow \frac{dy}{dx} = e^x\ \{2\ \cot 2x\ .sec^2(2x) + \log\tan 2x\}$$ Answer.

4. $If\ y = \tan^{-1}\left(\frac{2x}{1-x^2}\right), prove\ that\ \frac{dy}{dx} = \frac{2}{1+x^2}.$ $(isc\ 2003)$

Solution: Given $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$$= \tan^{-1}\left(\frac{2\tan\theta}{1-tan^2\theta}\right) \qquad Let\ x = \tan\theta$$

$$= \tan^{-1}(\tan 2\theta) \qquad \therefore\ \theta = \tan^{-1} x$$

$$= 2\theta$$

$$= 2\tan^{-1} x$$

$$\therefore\ \frac{dy}{dx} = \frac{2}{1+x^2}$$ Proved.

5. $If\ y = e^{m\cos^{-1}x}, prove\ that\ (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2 y$ $(isc\ 2003)$

Solution: Given $y = e^{m\cos^{-1}x}$

Taking log both sides

$$\Rightarrow \quad \log y = m\cos^{-1} x$$

$Differentiating\ each\ term\ w.r.t.x$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = -\frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \sqrt{1-x^2}\frac{dy}{dx} = -m\,y$$

$Squaring\ both\ sides$

$$\Rightarrow \quad (1-x^2)\left(\frac{dy}{dx}\right)^2 = m^2 y^2$$

$Differentiating\ each\ term\ w.r.t.\ x$

$$\Rightarrow \quad (1-x^2)2\frac{dy}{dx}\frac{d^2y}{dx^2} + (-2x)\left(\frac{dy}{dx}\right)^2 = m^2\,2y\frac{dy}{dx}$$

$Dividing\ each\ term\ by\ 2\frac{dy}{dx}$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2 y$$ Proved

6. $If\ x^y y^x = 5, show\ that\ \frac{dy}{dx} = -\left(\frac{\log y + \frac{y}{x}}{\log x + \frac{x}{y}}\right)$ $(isc\ 2004)$

Solution: Given $x^y y^x = 5$

Taking log both sides

$$\Rightarrow \quad \log x^y + \log y^x = \log 5$$

$$\Rightarrow \quad y\log x + x\log y = \log 5$$

$Differentiating\ each\ term\ w.r.t.\ x$

$$\Rightarrow \quad y\frac{1}{x} + \log x\frac{dy}{dx} + \log y + x\frac{1}{y}\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \left(\log x + \frac{x}{y}\right)\frac{dy}{dx} = -\frac{y}{x} - \log y$$

$$\Rightarrow \quad \frac{dy}{dx} = -\left(\frac{\log y + \frac{y}{x}}{\log x + \frac{x}{y}}\right)$$ Proved

7. $If\ x = a\,sin^3 t\ and\ y = a\,cos^3 t,\ find\ \frac{dy}{dx}.$ $(isc\ 2004)$

Solution: Given $x = a\,sin^3 t$ | $y = a\,cos^3 t$

Differentiating $w.r.t.\ t$

$$\Rightarrow \quad \frac{dx}{dt} = a.3sin^2 t.\cos t \qquad \Rightarrow \quad \frac{dy}{dt} = a\,3\,cos^2 t.(-\sin t)$$

$\Rightarrow \quad \frac{dx}{dt} = 3\,a\,sin^2 t.\cos t \quad \Big| \quad \Rightarrow \quad \frac{dy}{dt} = -3a\,cos^2 t.\sin t$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-3a\,cos^2 t.\sin t}{3\,a\,sin^2 t.\cos t}$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{\cos t}{\sin t}$$

$$\Rightarrow \quad \frac{dy}{dx} = -\cot t$$ Answer.

8. *If* $\sin(xy) + \cos(xy) = 1$ *and* $\tan(xy) \neq 1$, *then show that* $\frac{dy}{dx} = -\frac{y}{x}$. (*isc* 2005)

Solution: Given $\sin(xy) + \cos(xy) = 1$

Differentiating $w.r.t.\ x$

$$\Rightarrow \quad \cos(xy)\left(y + x\frac{dy}{dx}\right) - \sin(xy)\left(y + x\frac{dy}{dx}\right) = 0$$

$$\Rightarrow \quad \cos(xy).x\frac{dy}{dx} - \sin(xy).x\frac{dy}{dx} = -\cos(xy).y + \sin(xy).y$$

$$\Rightarrow \quad \{\cos(xy) - \sin(xy)\}\,x\frac{dy}{dx} = y\,\{\sin(xy) - \cos(xy)\}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y\,\{\sin(xy) - \cos(xy)\}}{x\,\{\cos(xy) - \sin(xy)\}}$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{y}{x}$$ Proved.

9. *If* $x^p\,y^q = (x+y)^{p+q}$, *prove that* $\frac{dy}{dx} = \frac{y}{x}$. (*isc* 2005)

Solution: Given $x^p\,y^q = (x+y)^{p+q}$

Taking log both sides

$$\Rightarrow \quad p\,\log x + q\,\log y = (p+q)\log(x+y)$$

Differentiating each term w.r.t. x

$$\Rightarrow \quad p.\frac{1}{x} + q\,\frac{1}{y}.\frac{dy}{dx} = (p+q)\frac{1}{x+y}.\left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \quad \frac{p}{x} + \frac{q}{y}.\frac{dy}{dx} = \frac{p+q}{x+y} + \frac{p+q}{x+y}.\frac{dy}{dx}$$

$$\Rightarrow \quad \left(\frac{q}{y} - \frac{p+q}{x+y}\right)\frac{dy}{dx} = \frac{p+q}{x+y} - \frac{p}{x}$$

$$\Rightarrow \quad \left(\frac{qx+qy-py-qy}{y(x+y)}\right)\frac{dy}{dx} = \frac{px+qx-px-py}{x\,(x+y)}$$

$$\Rightarrow \quad \frac{qx-py}{y\,(x+y)}.\frac{dy}{dx} = \frac{qx-py}{x\,(x+y)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(qx-py)\;y\;(x+y)}{x\,(x+y)\,(qx-py)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{x}$$ Proved

10. *If* $y = e^{\sin x^2}$, *find* $\frac{dy}{dx}$. (*isc* 2006)

Solution: Given $y = e^{\sin x^2}$

Taking log both sides

$$\Rightarrow \quad \log y = \sin x^2$$

Differentiating each term w.r.t. x

$$\Rightarrow \quad \frac{1}{y}.\frac{dy}{dx} = \cos x^2.2x$$

$$\Rightarrow \quad \frac{dy}{dx} = y.2x\,\cos x^2$$

$$\Rightarrow \quad \frac{dy}{dx} = 2x.e^{\sin x^2}.\cos x^2$$ Answer.

11. $If\ y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}, prove\ that\ (1-x^2)\frac{dy}{dx} - xy = 1$ $(isc\ 2006)$

Solution: Given $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$

$$\Rightarrow \quad \sqrt{1-x^2}.y = \sin^{-1} x$$

$$\Rightarrow \quad \sqrt{1-x^2}\frac{dy}{dx} + \frac{1}{2\sqrt{1-x^2}}(-2x)y = \frac{1}{\sqrt{1-x^2}}$$ (differentiating each term $w.r.t.\ x$)

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} - x\,y = 1$$ Proved.

12. $If\ e^{x+y} = xy, show\ that\ \frac{dy}{dx} = \frac{y\,(1-x)}{x\,(1-y)}$ $(isc\ 2007)$

Solution: Given $e^{x+y} = xy$

Taking log both sides

$$\Rightarrow \quad x + y = \log x + \log y$$

$Differentiating\ each\ term\ w.r.t.\ x$

$$\Rightarrow \quad 1 + \frac{dy}{dx} = \frac{1}{x} + \frac{1}{y}.\frac{dy}{dx}$$

$$\Rightarrow \quad \left(1 - \frac{1}{y}\right)\frac{dy}{dx} = \frac{1}{x} - 1$$

$$\Rightarrow \quad \left(\frac{y-1}{y}\right)\frac{dy}{dx} = \frac{1-x}{x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y\,(1-x)}{x\,(y-1)}$$ Proved

13. $If\quad \sin y = x\ \sin(a+y), show\ that\ \frac{dy}{dx} = \frac{sin^2(a+y)}{\sin a}$ $(isc\ 2007)$

Solution : Given $\sin y = x\ \sin(a+y)$

$$\Rightarrow \quad x = \frac{\sin y}{\sin(a+y)}$$

Differentiating both sides $w.r.t.\ y$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\sin(a+y)\ \cos y - \cos(a+y)\ \sin y}{sin^2(a+y)}$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\sin(a+y-y)}{sin^2(a+y)}$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\sin a}{sin^2(a+y)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{sin^2(a+y)}{\sin a}$$ Proved

14. $Find\ \frac{dy}{dx}\quad if\quad y = \tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ $(isc\ 2008)$

Solution: Given $y = \tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ $\qquad Let\ x = \tan\theta$

$$= \tan^{-1}\frac{\sqrt{1+tan^2\theta}-1}{\tan\theta} \qquad \therefore\ \theta = \tan^{-1} x$$

$$= \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$= \tan^{-1}\left(\frac{2\,sin^2\frac{\theta}{2}}{2\,\sin\frac{\theta}{2}\,\cos\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \frac{\theta}{2}$$

$$= \frac{1}{2}\tan^{-1}x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\,(1+x^2)}$$ Answer

15. $If\ y = \sqrt{\frac{1-\cos x}{1+\cos x}}, find\ \frac{dy}{dx}$ $(isc\ 2008)$

Solution: Given $y = \sqrt{\frac{1-\cos x}{1+\cos x}}$

$$= \sqrt{tan^2\left(\frac{x}{2}\right)}$$

$$= \tan\left(\frac{x}{2}\right)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2}\,sec^2\left(\frac{x}{2}\right)$$ Answer.

16. Using a suitable substitution, find the derivative of $\tan^{-1}\frac{4\sqrt{x}}{1-4x}$ $(isc\ \ 2009)$

Solution: Let $y = \tan^{-1}\frac{4\sqrt{x}}{1-4x}$ $\qquad Let\ 2\sqrt{x} = \tan\theta$

$$= \tan^{-1}\frac{2.2\sqrt{x}}{1-(2\sqrt{x})^2} \qquad \therefore\ \theta = \tan^{-1}(2\sqrt{x})$$

$$= \tan^{-1}\left(\frac{2\,\tan\theta}{1-tan^2\theta}\right)$$

$$= \tan^{-1}(\tan 2\theta)$$

$$= 2\,\theta$$

$$= 2\,\tan^{-1}(2\sqrt{x})$$

$$\therefore \frac{dy}{dx} = \frac{2}{1+(2\sqrt{x})^2}\,.2\,.\frac{1}{2\sqrt{x}}$$

$$= \frac{2}{\sqrt{x}\,(1+4x)}$$ Answer.

17. Find the derivative of $\sin x^2$ with respect to x^3. $(isc\ 2009)$

Solution: $Let\ \ y = \sin x^2 \qquad and\ \ z = x^3$

$$\Rightarrow \quad \frac{dy}{dx} = \cos(x^2).2x \qquad and \quad \frac{dz}{dx} = 3\,x^2$$

$$\therefore \frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{2\,x\,\cos x^2}{3\,x^2}$$

$$= \frac{2\,\cos x^2}{3\,x} \qquad \text{Answer.}$$

18. Using a suitable substitution, find the derivative of $\tan^{-1}\sqrt{\frac{a-x}{a+x}}$ with respect to x. $(isc\ 2010)$

Solution: $Let \quad y = \tan^{-1}\sqrt{\frac{a-x}{a+x}}$ $\qquad Let\ x = a\,\cos\theta$

$$= \tan^{-1}\sqrt{\frac{a-a\,\cos\theta}{a+a\,\cos\theta}} \qquad \therefore\ \theta = \cos^{-1}\left(\frac{x}{a}\right)$$

$$= \tan^{-1}\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

$$= \tan^{-1}\sqrt{tan^2\left(\frac{\theta}{2}\right)}$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \frac{\theta}{2}$$

$$= \frac{1}{2}\cos^{-1}\left(\frac{x}{a}\right)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2}.\frac{-1}{\sqrt{1-\left(\frac{x}{a}\right)^2}}.\frac{1}{a}$$

$$= -\frac{1}{2\sqrt{a^2-x^2}} \qquad \text{Answer}$$

19. $If\ \ y = x^x,\ \ prove\ that\ \ \frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$ $\qquad (isc\ 2011)$

Solution : Given $\quad y = x^x$

Taking log both sides

$$\Rightarrow \quad \log y = x\ \log x$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = x.\frac{1}{x} + \log x \qquad \text{(differentiating each term } w.r.t.\ x)$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = 1 + \log x$$

$$\Rightarrow \quad \frac{1}{y}\frac{d^2y}{dx^2} + \frac{dy}{dx}.\left(-\frac{1}{y^2}\right)\frac{dy}{dx} = \frac{1}{x} \qquad \text{(differentiating again } w.r.t.x)$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0 \qquad \text{Proved.}$$

20. $If \quad e^y(x+1) = 1, then\ show\ that\ \frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2.$ $(isc\ 2012)$

Solution: Given $e^y(x+1) = 1$

$$\Rightarrow \quad e^y = \frac{1}{x+1}$$

$$\Rightarrow \quad y = \log\left(\frac{1}{x+1}\right)$$

$$\Rightarrow \quad y = -\log(x+1)$$

$$\therefore \quad \frac{dy}{dx} = -\frac{1}{x+1} \quad \text{and} \quad \left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{x+1}\right)^2 \quad \text{---------- (1)}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = -\frac{(-1)}{(x+1)^2}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \left(\frac{1}{x+1}\right)^2$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$$ [comparing with equation (1)] Proved

21. $If\ \ y = (\cot^{-1}x)^2, show\ that\ \ (1+x^2)^2\frac{d^2y}{dx^2} + 2x(1+x^2)\frac{dy}{dx} = 2$ $(isc\ 2013)$

Solution: Given $y = (\cot^{-1}x)^2$

Differentiating $w.r.t.\ x$

$$\Rightarrow \quad \frac{dy}{dx} = 2\cot^{-1}x.\left(\frac{-1}{1+x^2}\right)$$

$$\Rightarrow \quad (1+x^2)\frac{dy}{dx} = -2\cot^{-1}x$$

$Differentiating\ again\ w.r.t.\ x$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = \frac{2}{1+x^2}$$

$$\Rightarrow \quad (1+x^2)^2\frac{d^2y}{dx^2} + 2x(1+x^2)\frac{dy}{dx} = 2$$ Proved.

22. $If\ \ y = \frac{x\sin^{-1}x}{\sqrt{1-x^2}}, prove\ that\ \ (1-x^2)\frac{dy}{dx} = x + \frac{y}{x}$ $(isc\ 2014)$

Solution: Given $y = \frac{x\sin^{-1}x}{\sqrt{1-x^2}}$ ---------- (1)

$$\Rightarrow \quad \sqrt{1-x^2}.y = x\sin^{-1}x$$

$Differentiating\ w.r.t.\ x$

$$\Rightarrow \quad \sqrt{1-x^2}\frac{dy}{dx} + \frac{1}{2}.\frac{1}{\sqrt{1-x^2}}.(-2x)y = x.\frac{1}{\sqrt{1-x^2}} + \sin^{-1}x$$

$multiplying\ each\ term\ by\ \ \sqrt{1-x^2}$

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} - xy = x + \sin^{-1}x.\sqrt{1-x^2}$$

[putting value of $\sin^{-1}x$ from (1)]

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} - xy = x + \frac{y\sqrt{1-x^2}}{x}.\sqrt{1-x^2}$$

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} - xy = x + \frac{y(1-x^2)}{x}$$

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} - xy = x + \frac{y}{x} - yx$$

$$\Rightarrow \quad (1-x^2)\frac{dy}{dx} = x + \frac{y}{x}$$ Proved.

23. $If\ \ y = e^{m\cos^{-1}x}, prove\ that\ \ (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m^2y$ $(isc\ 2015)$

Solution: Given $y = e^{m\cos^{-1}x}$

Taking log both sides

$\log y = m \cos^{-1} x$

Differentiating $w.r.t.\ x$

$\frac{1}{y}\frac{dy}{dx} = m\left(-\frac{1}{\sqrt{1-x^2}}\right)$

$\Rightarrow \quad \sqrt{1-x^2}\,\frac{dy}{dx} = -m\,y$

$squaring\ both\ sides$

$\Rightarrow \quad (1-x^2)\left(\frac{dy}{dx}\right)^2 = m^2\,y^2$

$Differentiating\ again\ w.r.t.\ x$

$\Rightarrow \quad (1-x^2)2\,\frac{dy}{dx}\frac{d^2y}{dx^2} + (-2x)\left(\frac{dy}{dx}\right)^2 = m^2.2y\,\frac{dy}{dx}$

$dividing\ \ each\ term\ by\ \ 2\,\frac{dy}{dx}$

$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\,\frac{dy}{dx} = m^2\,y$ Proved.

24. $If\ \ \log y = \tan^{-1} x\ ,prove\ that\ \ (1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$ $(isc\ 2016)$

Solution: Given $\log y = \tan^{-1} x$

Differentiating both sides $w.r.t.\ x$

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{1+x^2}$

$\Rightarrow \quad (1+x^2)\frac{dy}{dx} = y$

$Differentiating\ again\ \ w.r.t\ \ x$

$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2} + 2x\,\frac{dy}{dx} = \frac{dy}{dx}$

$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$ Proved

25. $If\ \ y = \cos(\sin x)\,,show\ that\ \ \frac{d^2y}{dx^2} + \tan x\,\frac{dy}{dx} + y\,cos^2x = 0$ $(isc\ \ 2017)$

Solution: Given $y = \cos(\sin x)$

Differentiating both sides $w.r.t.\ x$

$\Rightarrow \quad \frac{dy}{dx} = -\sin(\sin x).\cos x$

$again\ differentiating\ both\ sides\ w.r.t.\ x$

$\Rightarrow \quad \frac{d^2y}{dx^2} = -\cos(\sin x).\cos x.\cos x - \sin(\sin x).(-\sin x)$

$\Rightarrow \quad \frac{d^2y}{dx^2} = -\cos(\sin x).cos^2x + \sin(\sin x).\sin x$

$Now\ \ L.H.S = \frac{d^2y}{dx^2} + \tan x\,\frac{dy}{dx} + y\,cos^2x$

$= -\cos(\sin x).cos^2x + \sin(\sin x).\sin x + \frac{\sin x}{\cos x}.\{-\sin(\sin x).\cos x\} + \cos(\sin x).cos^2x$

$= -\cos(\sin x).cos^2x + \sin(\sin x).\sin x - \sin x.\sin(\sin x) + \cos(\sin x).cos^2x$

$= 0$ Proved

26. $If\ \ x = \tan\left(\frac{1}{a}\log y\right)$ $, prove\ that\ \ (1+x^2)\frac{d^2y}{dx^2} + (2x-a)\frac{dy}{dx} = 0$ $\quad (isc\ \ 2018)$

Solution: Given $\quad x = \tan\left(\frac{1}{a}\log y\right)$

$\Rightarrow \quad \tan^{-1}x = \frac{1}{a}\log y$

$\Rightarrow \quad \log y = a\ \tan^{-1}x$

Differentiating both sides $w.r.t.\ x$

$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = a\ \frac{1}{1+x^2}$

$\Rightarrow \quad (1+x^2)\frac{dy}{dx} = a\,y$

$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = a\ \frac{dy}{dx}$ (Differentiating each term $w.r.t.x$)

$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2} + (2x-a)\ \frac{dy}{dx} = 0$ Proved

27. $If\ \ y = e^{\sin^{-1}x}\ and\ \ z = e^{-\cos^{-1}x}, prove\ that\ \ \frac{dy}{dz} = e^{\pi/2}$ $\quad (isc\ 2019)$

Solution: $y = e^{\sin^{-1}x}$ and $z = e^{-\cos^{-1}x}$

$\Rightarrow \quad \frac{dy}{dx} = e^{\sin^{-1}x}.\frac{1}{\sqrt{1-x^2}}$ and $\frac{dz}{dx} = e^{-\cos^{-1}x}.\left(-\frac{1}{-\sqrt{1-x^2}}\right)$

$$\therefore \quad \frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{e^{\sin^{-1}x}.\frac{1}{\sqrt{1-x^2}}}{e^{-\cos^{-1}x}.\frac{1}{\sqrt{1-x^2}}}$$

$$= \frac{e^{\sin^{-1}x}}{e^{-\cos^{-1}x}}$$

$$= e^{\sin^{-1}x+\cos^{-1}x}$$

$$= e^{\pi/2}$$ Proved

28. $Find\ \ \frac{dy}{dx}\ \ if\ \ x^3+y^3 = 3axy$ $\quad (isc\ 2020)$

Solution: Given $\quad x^3+y^3 = 3axy$

Differentiating both sides $w.r.t.x$,

$$3x^2 + 3y^2\ \frac{dy}{dx} = 3a\left(x\frac{dy}{dx} + y\right)$$

$\Rightarrow \quad x^2 + y^2\frac{dy}{dx} = ax\ \frac{dy}{dx} + ay$

$\Rightarrow \quad (y^2 - ax)\frac{dy}{dx} = ay - x^2$

$\Rightarrow \quad \frac{dy}{dx} = \frac{ay-x^2}{y^2-ax}$ Answer

29. $If\ \ y = e^{m\ \sin^{-1}x},\ prove\ \ that\ \ (1-x^2)\frac{d^2y}{dx^2} - x\ \frac{dy}{dx} = m^2\,y$ $\quad (isc\ \ 2020)$

Solution: Given $\quad y = e^{m\ \sin^{-1}x}$

Taking log on both sides

$\Rightarrow \quad \log y = m\ \sin^{-1}x$

Differentiating each term $w.r.t.x$

$\Rightarrow \quad \frac{1}{y}\ \frac{dy}{dx} = \frac{m}{\sqrt{1-x^2}} \qquad \Rightarrow \quad \sqrt{1-x^2}\ \frac{dy}{dx} = m\,y$

$\Rightarrow \quad (1-x^2)\left(\frac{dy}{dx}\right)^2 = m^2\,y^2$ (squaring both sides)

Differentiating again each term $w.r.t.x$

$$\Rightarrow \quad (1-x^2)2\,\frac{dy}{dx}\,\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = m^2.2y\,\frac{dy}{dx}$$

Dividing each term by $2\,\frac{dy}{dx}$,

$$\Rightarrow \quad (1-x^2)\,\frac{d^2y}{dx^2} - x\,\frac{dy}{dx} = m^2\,y$$ Proved

30. Select and write the correct option: $(isc\ Sem\ 1-2021)$

Differentiation of $\log(1+x^2)$ with respect to $\tan^{-1}x$ is:

(a) $\frac{1}{1+x^2}$
(b) $2x$
(c) $\frac{-1}{1+x^2}$
(d) $-2x$

Answer: _____

Solution: (b)

Let $y = \log(1+x^2)$ and $z = \tan^{-1}x$

$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^2} \times 2x$ $\qquad \Rightarrow \frac{dy}{dz} = \frac{1}{1+x^2}$

$$\therefore \frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{\frac{2x}{1+x^2}}{\frac{1}{1+x^2}} = 2x$$

31. Select and write the correct option: $(isc\ Sem\ 1-2021)$

If $x^2+y^3=42$, then $\frac{dy}{dx}$ is:

(a) $\frac{dy}{dx} = \frac{-3\,y^2}{2x}$
(b) $\frac{dy}{dx} = \frac{3\,y^2}{2\,x}$
(c) $\frac{dy}{dx} = \frac{2\,x}{3\,y^2}$
(d) $\frac{dy}{dx} = \frac{-2\,x}{3\,y^2}$

Answer: ----------

Solution: (d)

$x^2+y^3=42$

$\Rightarrow \quad 2x + 3\,y^2\,\frac{dy}{dx} = 0$ (Differentiating both sides)

$\Rightarrow \quad 3\,y^2\,\frac{dy}{dx} = -2x$

$\Rightarrow \quad \frac{dy}{dx} = \frac{-2x}{3\,y^2}$

32. Select and write the correct option: $(isc\ Sem\ 1 - 2021)$

If $x = a\ \cos\theta$, $y = a\ \sin\theta$, then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$ will be:

(a) $\frac{dy}{dx} = -1$

(b) $\frac{dy}{dx} = 1$

(c) $\frac{dy}{dx} = 0$

(d) $\frac{dy}{dx} = 2$

Answer: _____

Solution: (c)

$x = a\ \cos\theta \quad \Rightarrow \frac{dx}{d\theta} = -a\ \sin\theta$

$y = a\ \sin\theta \quad \Rightarrow \frac{dy}{d\theta} = a\ \cos\theta$

$\therefore\ \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a\ \cos\theta}{-a\ \sin\theta} = -\ \cot\theta$

$\therefore\ \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}} = -\ \cot\frac{\pi}{2} = 0$

33. Select and write the correct option: $(isc\ Sem\ 1 - 2021)$

If $y = \tan^{-1} x$, then

(a) $(1 + x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(b) $\sqrt{1 - x^2}\ \frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(c) $(1 - x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

(d) $\sqrt{1 + x^2}\ \frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$

Answer: _____

Solution: (a)

$y = \tan^{-1} x$ (Differentiating $w.r.t.x$)

$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^2} \quad \Rightarrow (1 + x^2)\frac{dy}{dx} = 1$

$\Rightarrow (1 + x^2)\frac{d^2y}{dx^2} + 2x\ \frac{dy}{dx} = 0$ (Again differentiating $w.r.t.x$)